Charagundla Hemanth Sai

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PROFESSIONAL SUMMARY

Artificial Intelligence & Data Science undergraduate with hands-on experience designing end-to-end data engineering, analytics, and machine learning solutions using PostgreSQL, Python, and Power BI for business decision support.

EDUCATION

B.E – Artificial Intelligence and Data Science Expected Graduation: 2028
Chaitanya Bharathi Institute of Technology (CBIT) CGPA: 8.73

SKILLS

Languages: Python | SQL | C++ | Java | R
Databases: PostgreSQL | Oracle SQL
Data Science & Machine Learning: Pandas | NumPy | Scikit-learn | XGBoost | TensorFlow | SHAP
Visualization: Power BI | Tableau | Plotly | Matplotlib | Seaborn
Tools: Git | GitHub | Streamlit | Jupyter Notebook | VS Code
Core Concepts: Statistics | Probability | Linear Algebra | Data Structures & Algorithms | OOP

PROJECTS

Healthcare Analytics & Business Intelligence System — PostgreSQL | SQL | Power BI

- Designed a production-style PostgreSQL healthcare warehouse integrating 16.3M+ SPARCS records across 7 years using a layered ETL architecture and Star Schema.
- Engineered a reusable multi-year ETL pipeline supporting 7 years of incremental loading with surrogate keys, Unknown Member handling, audit logging, data validation, and date dimension integration.
- Engineered an analytics layer comprising 11 SQL analytical views, 6 materialized views, 8 business reporting modules, and 8 interactive Power BI dashboards for executive, financial, clinical, operational, geographic, and trend analysis.
- Optimized analytical workloads using indexes, planner statistics, EXPLAIN ANALYZE, and materialized views while documenting the ETL architecture, KPIs, and business reporting workflow.

Medical Inventory Forecasting & Decision Support System — Python | Scikit-learn | SHAP | Streamlit

- Built an end-to-end Medical Inventory Forecasting & Decision Support System integrating data preprocessing, feature engineering, SHAP explainability, K-Means segmentation, and business decision support.
- Evaluated 14 machine learning and ensemble models, selecting a Gradient Boosting Regressor that achieved an R^2 of 0.7978, RMSE of 9.692, and MAE of 4.415 for demand forecasting.
- Implemented SHAP explainability, error analysis, and K-Means segmentation to identify demand patterns, interpret model predictions, and support procurement planning and inventory risk assessment.
- Designed and deployed a multi-page Streamlit application featuring interactive dashboards, demand prediction, SHAP explainability, inventory segmentation, and business recommendation modules.

ADDITIONAL PROJECTS

Anime & Manga Analytics Platform — Python | Pandas | NumPy | Matplotlib | Seaborn

- Conducted exploratory data analysis and implemented a content-based recommendation workflow to analyze genres, ratings, and viewing behavior.

Hate Speech Detection — Python | Scikit-learn | NLP

- Built a hate speech detection model using NLP preprocessing and supervised machine learning for text classification.

ACHIEVEMENTS

- Solved 120+ problems on LeetCode, including 80+ SQL and 35+ Data Structures & Algorithms problems.
- Earned HackerRank SQL (Advanced) certification.

LANGUAGES

English (Proficient) | Hindi (Proficient) | Telugu (Native)